

The Oriented Frontier Observable: a Recursive Q11 Diagnostic for the Angular Generation Split

(internal note, twin to the symmetric-capacity obstruction)

Cosmochrony programme — working note

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Abstract

The symmetric-capacity obstruction establishes that the raw shell observable of the angular generation split vanishes identically, $\Theta_{\text{raw}} = 0$: on a symmetric BFS shell the involution $g \mapsto g^{-1}$ pairs opposite central phases and erases the oriented J_{Π} -odd component before the radial recursion can act. This note defines the honest replacement: a frontier transfer observable attached to the directed outgoing edges $g \rightarrow gs$ of the cascade, whose central increment is the Heisenberg cocycle $a(g) s_b$, the discrete counterpart of the Baker–Campbell–Hausdorff term $\frac{1}{2}[X, Y]$ and hence of the dynamical J_3 [4, 3]. The orientation is the recursive arrow $\partial_\tau = \log g$ from the projection origin to increasing BFS depth, not an extracted weight. The first test is not an amplitude value but the existence of an oriented signal, $\langle \Delta A_c \rangle_{\partial^+ S_m} \neq 0$, which may fail if a residual automorphism re-pairs the frontier; the discriminating anti-bias control is the symmetrised-frontier cancellation $\langle \Delta A_c \rangle_{\partial^+ S_m \cup \partial^- S_m} = 0$. No normalisation $\mathcal{N}_A$, no dictionary value, and no ε appear in this note: it tests only whether the recursion produces a non-zero, bias-free oriented signal.

Keywords: oriented cascade; frontier transfer; Heisenberg cocycle; recursive orientation; generation split; anti-bias control; projected capacity.

1 Context

Q14, §6 [3] fixes the angular generation split as an amplitude with a factorised architecture and defers its absolute value to the physical normalisation of the metaplectic J_3 generator. The companion symmetric-capacity obstruction note isolates the two sectors in the O12 fingerprint exponent — the frequency B_c carries the radial capacity σ_c , the central phase A_c carries the angular split — and shows that a direct evaluation on symmetric BFS shells returns $\Theta_{\text{raw}} = 0$ identically. The obstruction is not the absence of an angular phase; it is that the symmetric shell average is the wrong observable for a recursively oriented cascade.

The present note records the natural exit, prescribed by the oriented cascade arrow $\partial_\tau = \log g$ [4] rather than chosen. Concretely, the central phase of a single element $g = (a, b, z)$ under the central character c is $A_c(g) = cz$, so the swept central coordinate z is the oriented symplectic area from the origin to g . Along one outward step $g \rightarrow gs$ the Heisenberg law gives $z(gs) = z(g) + z(s) + a(g) s_b$, hence

$$\Delta A_c(g, s) = A_c(gs) - A_c(g) = c(z(s) + a(g) s_b),$$

which for the admissible generators ($z(s) = 0$) reduces to the pure cocycle $ca(g) s_b$. This is the discrete $\frac{1}{2}[X, Y]$ increment: the transfer observable measures exactly the dynamical J_3 carried by the oriented edge, not a free shear.

2 The oriented frontier observable

The symmetric-capacity obstruction shows that the raw shell observable cannot select an angular branch: on a symmetric BFS shell S_m , the involution $g \mapsto g^{-1}$ pairs opposite central phases and cancels the oriented J_Π -odd component. The Q11-oriented observable is therefore not a shell average. It is a frontier transfer observable.

Fix the identity element e as the projection origin and let $d(e, g)$ be the BFS distance in the admissible Cayley graph. For each shell $S_m = \{g : d(e, g) = m\}$, define the outgoing and incoming frontiers by

$$\begin{aligned}\partial^+ S_m &= \{(g, s) : d(e, g) = m, d(e, gs) = m + 1\}, \\ \partial^- S_m &= \{(g, s) : d(e, g) = m, d(e, gs) = m - 1\}.\end{aligned}$$

No subset of generators is selected. The orientation is the recursive arrow from the projection origin to increasing BFS depth.

For a directed frontier edge $g \rightarrow gs$, define

$$\Delta A_c(g, s) = A_c(gs) - A_c(g).$$

Since A_c changes sign under the conjugate branch $c \mapsto q - c$, ΔA_c is already J_Π -odd:

$$\text{Odd}_{J_\Pi}(\Delta A_c) = \Delta A_c.$$

The J_Π -odd notation is therefore a parity check, not an additional filter.

The Q11-oriented raw angular profile is

$$\Theta_{\text{Q11}}^{\text{raw}}(m) = \left\langle \frac{2\pi}{q} \Delta A_c(g, s) \right\rangle_{\partial^+ S_m}.$$

Its cumulative, capacity-normalised form is

$$\Theta_{\text{Q11}}^{\text{oriented}}(n) = \frac{1}{\widehat{I}(n)} \sum_{m \leq n} \Theta_{\text{Q11}}^{\text{raw}}(m) \Delta \widehat{I}(m).$$

This quantity is prescribed by the oriented frontier of the cascade, not extracted from symmetric capacity data.

The first test is not the value of the amplitude, but the existence of an oriented signal:

$$\Theta_{\text{Q11}}^{\text{raw}}(m) \stackrel{?}{\neq} 0.$$

This test may fail. The outgoing frontier is canonical once the origin and admissible generators are fixed, but a residual automorphism of the frontier may still re-pair the cocycle contributions.

The required anti-bias control is the symmetrised-frontier cancellation:

$$\left\langle \frac{2\pi}{q} \Delta A_c(g, s) \right\rangle_{\partial^+ S_m \cup \partial^- S_m} \stackrel{!}{=} 0.$$

Thus a valid oriented signal must satisfy

$$\langle \Delta A_c \rangle_{\partial^+ S_m} \neq 0, \quad \langle \Delta A_c \rangle_{\partial^+ S_m \cup \partial^- S_m} = 0.$$

Only then can the signal be attributed to the recursive orientation rather than to a hidden sampling bias.

3 BI parity versus the residual reflection

The outgoing-frontier observable isolates a residual obstruction. The obstruction is not the BI parity itself. It is the reflection

$$\varphi : (a, b, z) \mapsto (-a, b, -z),$$

which preserves the outgoing frontier and reverses the central cocycle.

In the O12 fingerprint variables

$$\text{fp}_c(g; x) = q^{-3/2} \exp \left(\frac{2\pi i}{q} (A_c(g) + B_c(g)x) \right),$$

the action of φ is

$$\varphi : (A_c, B_c) \mapsto (-A_c, +B_c).$$

By contrast, the spinorial BI parity of Q14 [3] is the antiunitary structure $J_\Pi = HK$. In the weight basis Q14 fixes

$$K = \sigma_x \circ \overline{(\cdot)}, \quad H = \sigma_z, \quad J_\Pi^{\text{int}} = HK = \varepsilon \circ \overline{(\cdot)}.$$

Thus the induced action on the fingerprint character is the BI-conjugate action

$$J_\Pi : (A_c, B_c) \mapsto (-A_c, -B_c).$$

Consequently

$$J_\Pi \neq \varphi.$$

The residual reflection differs from BI parity by the frequency reflection

$$R_b : (A_c, B_c) \mapsto (A_c, -B_c),$$

namely

$$\varphi = R_b \circ J_\Pi.$$

This distinction is important. The residual obstruction is not removed by the statement that the chiral selector is J_Π -odd. Since φ differs from J_Π by R_b , the relevant condition is instead whether chirality is blind to the frequency sector:

$$[\gamma_5, R_b] = 0.$$

Under this condition, J_Π -oddness implies φ -oddness. Indeed, using $J_\Pi \gamma_5 = -\gamma_5 J_\Pi$ and assuming $[\gamma_5, R_b] = 0$, one obtains

$$\varphi \gamma_5 = R_b J_\Pi \gamma_5 = -R_b \gamma_5 J_\Pi = -\gamma_5 R_b J_\Pi = -\gamma_5 \varphi.$$

Thus γ_5 is φ -odd.

The frontier cocycle ΔA_c is itself φ -odd and arrow-odd. Therefore, if $[\gamma_5, R_b] = 0$, the chiral integrand

$$\gamma_5 \Delta A_c$$

is φ -even and arrow-odd. It can survive the residual reflection while retaining the required orientation under cascade reversal.

This is only a conditional lifting statement. If $[\gamma_5, R_b] \neq 0$, or if R_b is not a legitimate frequency-sector operator on the chiral carrier, then the $V - A$ selector of Q14 does not lift the φ -obstruction. In that case the angular amplitude remains open even after the Q11 frontier orientation is imposed.

4 Status and protocol

This note defines the object and its falsification protocol, and resolves the operator condition on which the chiral lift depends (Section 5). The twin script `q11_oriented_frontier.py` returns the three profiles $\Theta_{\partial^+}^{\text{raw}}(m)$, $\Theta_{\partial^-}^{\text{raw}}(m)$, $\Theta_{\partial^+ \cup \partial^-}^{\text{raw}}(m)$, the cumulative capacity-normalised $\Theta_{\text{Q11}}^{\text{oriented}}(n)$, and the three status flags (`sign_reversal_check`, `sym_frontier_zero_check`, `automorphism_residual_status`). The companion computation verifies this obstruction on $q \in \{61, 101, 151\}$: the outgoing-frontier signal satisfies $\max_m |\Theta_{\partial^+}^{\text{raw}}(m)| = 0$, while the residual reflection $\varphi : (a, b, z) \mapsto (-a, b, -z)$ preserves the outgoing frontier and reverses the cocycle ΔA_c . This numerical check is not used to define the observable; it only confirms the frontier re-pairing predicted by the symmetry. The sign-reversal check $\Theta_{\partial^-}^{\text{raw}} = -\Theta_{\partial^+}^{\text{raw}}$ is nearly tautological once ∂^- is the reversed frontier; it is a consistency check, not the discriminant. The discriminant is the symmetrised cancellation: a signal is admissible only if it is non-zero on ∂^+ and zero on $\partial^+ \cup \partial^-$.

The status of the observable is explicit: the orientation is prescribed by $\partial_\tau = \log g$, not extracted from symmetric capacity data. Neither the normalisation $\mathcal{N}_A$ nor any comparison to a dictionary value is performed here; these remain deferred, and become meaningful only if both tests pass.

The remaining problem is therefore not an amplitude computation. It is the operator question whether the chiral grading γ_5 commutes with the frequency reflection R_b . This question is settled in Section 5: the residual reflection is the central Weil lift $W(-I)$, hence a scalar on the chiral carrier. Only once this is established does it make sense to compute a chiral frontier signal and to ask for the normalisation $\mathcal{N}_A$.

5 Resolution: the residual reflection is the central Weil lift

The conditional lifting statement of Section 3 reduces the obstruction to a single operator question. The O12 fingerprint settles it: the residual frequency reflection is not a free operator but the Weil lift of the central symplectic element, so its action on the chiral carrier is forced.

Lemma 1 (Residual frequency reflection as the central Weil lift). *The residual reflection $R_b = \varphi \circ J_\Pi^{-1}$ is the Weil lift of the central symplectic element $-I$. Equivalently $R_b = W(-I) = \text{FT}^2$, up to the metaplectic phase.*

Proof. Both φ and J_Π invert the central character, $A_c \mapsto -A_c$, so each is antiunitary. Their quotient $R_b = \varphi \circ J_\Pi^{-1}$ is therefore unitary and preserves the central character. By the Stone–von Neumann theorem it is the Weil lift $W(M)$ of a symplectic automorphism M , up to phase. The element M is fixed by the fingerprint. The O12 fingerprint [1] has the form

$$\text{fp}_c(g; x) = q^{-3/2} \exp\left(\frac{2\pi i}{q}(A_c(g) + B_c(g)x)\right), \quad B_c(g) = \sum_i c_i b_i, \quad A_c(g) = \sum_i c_i(\gamma_i - b_i a_i),$$

so B_c is the pure Weil frequency carried by the b_i , while the cocycle $-b_i a_i$ sits in A_c . The action $R_b : (A_c, B_c) \mapsto (A_c, -B_c)$ forces $b_i \mapsto -b_i$. Invariance of A_c , including its cocycle term $-b_i a_i$, then forces $a_i \mapsto -a_i$, while the centre is fixed. Hence the induced horizontal map is $(a_i, b_i) \mapsto (-a_i, -b_i)$, that is $M = -I$, and $R_b = W(-I) = \text{FT}^2$ up to the metaplectic phase. $\square$

Corollary 1 (Commutation with chirality). *The residual frequency reflection satisfies $[\gamma_5, R_b] = 0$.*

Proof. By Lemma 1, R_b is the Weil lift of the central element $-I \in \text{Sp}(2)$. After the Lorentzian complexification $\mathfrak{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$ of Q14 [3], this element lies in the connected spin group $\text{SL}(2, \mathbb{C})$, where $-I$ is central and acts on each Weyl factor by a scalar. The metaplectic phase ambiguity

changes this scalar by an overall sign or phase only. In all cases R_b is a scalar on $S_L \oplus S_R$, while γ_5 is diagonal with opposite eigenvalues on the two factors. A scalar commutes with a diagonal operator, hence $[\gamma_5, R_b] = 0$. $\square$

The conditional lift of Section 3 is therefore unconditional on the chiral side. The residual reflection is blind to chirality, so the $V - A$ selector of Q14 [3] removes the φ -paired obstruction, and it does so without introducing a spacetime parity: R_b is the central element of the connected spin group, not a disconnected element of $O(3, 1)$. What remains deferred is only the amplitude normalisation $\mathcal{N}_A$, not the operator question.

The chiral weight γ_5 on which this signal depends is identified in the companion note [2]: it is not a function of the cascade generator $T(e)$ — arrow inversion and the residual reflection φ both act as $T \mapsto -T$, so every $f(T)$ has equal arrow- and φ -parity, whereas the admissible weight is arrow-even and φ -odd — but the Weyl representation type $\mathbf{2}$ versus $\bar{\mathbf{2}}$, which group inversion preserves and complex conjugation exchanges. In the inherited Heisenberg central-phase lift the edge phase is the linear cyclotomic character $\zeta_q^{\Delta A_c}$, with no Gauss, Maslov, or $\sqrt{5}$ component, and the outgoing-frontier overlap is maximal, $\rho_\chi = 1$, promoting the bound to $\mathcal{N}_A = \theta_{\max} \neq 0$ in that lift. The residual open point is an additional spin-Galois phase of the full Lorentzian chiral lift, in particular a $\sqrt{5}$ or ADE-spin factor orthogonal to ζ_q ; the spectral split $u \neq 0$ still requires the Schur transport [5, 2].

References

- [1] J. Beau. Exact Weil-Block Projective Capacity on Heisenberg Graphs: Resolving the Final Obstruction to δ Extraction, 2026. Spectral admissibility sub-program "O12" paper, doi:10.5281/zenodo.19194798.
- [2] Jérôme Beau. Chiral Orientation of the Directed Heisenberg Frontier, 2026. "CHO" note, doi:10.5281/zenodo.20735907.
- [3] Jérôme Beau. Fermionic Matter and Chirality from Projective Dirac Admissibility, 2026. "Q14" paper, doi:10.5281/zenodo.20218409.
- [4] Jérôme Beau. Temporal Casimir Rigidity and Closure of the Effective Metric: Identification of $A_\tau = 2$, 2026. "Q11" paper, doi:10.5281/zenodo.20098387.
- [5] Jérôme Beau. The Projective Residue as a Schur Complement: Reduction of the Generation Split to the Chiral Asymmetry of Projection Locking, 2026. Fermionic matter sub-programme note: exhibits the projective endomorphism as the Schur complement $E_\Pi = -\Pi_S D(1 - P) D \Pi_S^* = -M^\dagger M$ of the spinorial directions eliminated by the non-injective projection; reduces the inter-generation split coefficient u to the $\mathcal{D}^\pm$ -transported, generation-projected component of the antiunitary chiral equivariance defect $\Delta_\chi(P) = \pi_{LL} - \tau \overline{\pi_{RR}} \tau^{-1}$; stratifies the result on the Cosmochrony axioms (A1–A3 force $u = 0$; non-zero u requires A4 chiral symmetry-breaking); proves finite/Lorentzian separation (finite locking is J_Π -equivariant, so $u_{\text{fin}} = 0$); locks Seeley–DeWitt conventions so operator and spectral u coincide; no-go result: chirality is Lorentzian, not a finite-fibre label. Reduces the open problem to a Lorentzian computation of the chiral curvature $\mu_\chi^2 := \partial_s^2 \mathcal{B}(0)$ along the J_Π -odd modulus of the saturated Born–Infeld functional, doi:10.5281/zenodo.20601040.